

# **Hands-On Museum Modular Roller Coaster: Kinetics Exhibit**

DESCI 503 Project Report

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# Executive Summary

## Problem Statement/Improvement Opportunity

The Hands-On Museum currently lacks a hands-on exhibit that teaches visitors about conservation of mechanical energy and conservation of momentum in a fun and engaging way.

## Summary of Approach

The proposed exhibit is a modular roller coaster track that allows visitors to experiment with potential and kinetic energy, momentum, and friction. The track will be made up of several different sections that can be easily assembled and disassembled. Visitors will be able to select the number of cars per train to see how they affect the coaster's performance. There will also be informational displays that explain the scientific concepts involved.

## Estimated Financial Benefit

The cost of developing and implementing the exhibit is estimated to be \$10,000. The exhibit is expected to generate additional revenue for the museum through increased ticket sales and gift shop sales. Additionally, the museum plans to produce multiple copies of the exhibit to sell them for a profit. Each modular roller coaster exhibit will sell for \$15,000.

## Education Focus:

- **Potential and Kinetic Energy:** Through interactive track configurations, visitors will observe the conversion of potential energy (stored at height) into kinetic energy (motion) as the coaster car descends.
- **Momentum:** By experimenting with balls of varying masses, visitors will explore how mass impacts the coaster's movement and momentum.
- **Friction:** Interactive stations will allow users to test balls with different textures, demonstrating the effect of friction on speed and overall performance.

## Exhibit Components:

- **Modular Track System:** The centerpiece comprises 6-8 lightweight, easily assembled track sections featuring various elements like hills, twists, loops, and an initial drop hill (pre-installed). These modular pieces will connect seamlessly via a puzzle piece design, allowing for a vast array of customizable coaster configurations.
- **Informational Displays:** Educational displays will be integrated along the track, visually demonstrating the transfer of potential and kinetic energy. Informative signage will be present throughout, explaining concepts like momentum and friction in an age-appropriate manner.
- **Hands-On Learning Stations:** Visitors can select the number of cars they attach to their train. This allows them to observe how changes in mass affect the coaster's ability to navigate the track and highlights the influence of these factors on the core physics principles.

- **Problem-Solving Challenges:** Interactive challenges will encourage visitors to apply their newfound knowledge. Predetermined track configurations can be provided, prompting users to solve problems and find successful configurations to complete the course. These challenges will cater to various skill levels, ensuring continuous engagement for all ages.

### Benefits and Impact:

- **Educational Value:** The exhibit offers a dynamic and immersive learning experience, fostering curiosity and interest in physics, engineering, and science through hands-on exploration.
- **Engaging Experience:** The interactive nature and customizable track system promote exploration and experimentation, making learning fun and memorable for all visitors.
- **STEM Education Tool:** The exhibit serves as a valuable resource for educators, providing a platform to introduce STEM concepts with a hands-on approach.
- **Museum Enhancement:** This visually engaging and thematically cohesive exhibit aligns with the museum's mission while attracting visitors and enhancing their overall experience.

## 1.0 Introduction and Current State

The Hands-On Museum seeks to enhance its educational offerings by developing a new interactive exhibit focused on teaching fundamental physics principles through hands-on learning. The purpose of this report is to document the process of conceptualizing, designing, prototyping, and testing an engaging and educational exhibit that will help visitors, particularly children, understand concepts such as potential and kinetic energy, momentum, and friction. The scope of this project encompasses the entire development cycle, from initial ideation to the creation of a testable prototype.

### *1.1 Overview of the Project*

The project involves creating a modular roller coaster exhibit that will be a part of the Hands-On Museum's array of interactive learning experiences. This exhibit will feature components that visitors can assemble to form different roller coaster tracks, allowing them to explore the effects of various design elements on the coaster's performance. The interactive nature of the exhibit aims to foster curiosity and inspire interest in physics and engineering among visitors.

The project is situated within the museum and educational sector, specifically focusing on informal science education. The Ann Arbor Hands-On Museum, known for its interactive exhibits, serves as a hub for learning and exploration for children and families. The social context of the project emphasizes the importance of engaging educational experiences that make learning enjoyable and accessible.

Key stakeholders in this project include:

- Museum Visitors: Primarily children aged 3-12 and their families who will interact with the exhibit.
- Educators: Teachers and museum staff who will use the exhibit as a tool to teach physics concepts.
- Museum Administration: The team responsible for curating and maintaining exhibits.
- Project Team: Individuals involved in the design, development, and implementation of the exhibit.

This topic was selected to address the need for more interactive and educational exhibits at the museum that can engage a diverse audience and provide hands-on learning opportunities. The potential benefits of this project include:

- Enhanced Learning: Visitors will gain a better understanding of physics mechanics and kinetics principles through practical, interactive experiences.
- Increased Engagement: The exhibit's interactive nature will likely increase visitor engagement and retention.
- Educational Support: The exhibit can serve as a valuable resource for educators to teach STEM concepts in an engaging manner.
- Visitor Attraction: A new, innovative exhibit can attract more visitors to the museum, boosting attendance and revenue.

The critical questions guiding this project include:

- How can we design an exhibit that effectively teaches the principles of potential and kinetic energy, momentum, and friction?
- What design elements are necessary to ensure the exhibit is engaging and educational for a wide age range?
- How can we ensure the safety and durability of the exhibit components?

These questions are important because they address the core objectives of the project: to educate, inspire, and entertain visitors while providing a safe and sustainable learning environment. The problem at hand is the need for an engaging educational exhibit that can simplify complex physics concepts and make them accessible to young learners. This project is important because it will fill a gap in the museum's current offerings and contribute to the overall mission of providing hands-on, educational experiences.

## 1.2 Review of Current State Design / Literature Review

### Summary Prior Work

The project builds on a legacy of interactive exhibits designed to educate visitors about scientific principles through engaging and hands-on experiences. Previous projects at the Hands-On Museum have successfully implemented various kinetic educational exhibits, such as the "Window Maze" exhibit and the "Engineers On a Roll" exhibit. These exhibits provide a solid foundation and valuable insights into designing educational content that visually communicates concepts of physics mechanics in an accessible manner.



Figure 1: Window Maze (left), Engineers On a Roll (right) Exhibits

### Benchmarking

To benchmark the proposed modular roller coaster exhibit, a comprehensive analysis was conducted on existing state-of-the-art interactive exhibits in similar institutions. The benchmark included detailed evaluations of exhibits at prominent science museums renowned for their innovation in exhibit design and educational impact. This analysis helps to identify best practices and features that could be integrated into the proposed project to enhance its effectiveness and appeal.

## Exploratorium in San Francisco

### Exhibit: Tinkering Studio



Figure 2: Tinkering Studio Exhibit at the Exploratorium in San Francisco

The Tinkering Studio is an interactive space where visitors engage in hands-on activities to explore mechanical concepts, engineering, and physics. It emphasizes creativity and experimentation, encouraging visitors to build, test, and modify their creations. Visitors use various materials and tools to create mechanisms and structures, fostering a deep understanding of mechanical principles through trial and error. The exhibit features modular components that can be easily assembled and reconfigured, allowing for endless experimentation and learning opportunities. Clear explanations and prompts guide visitors through the learning process, making complex concepts accessible and engaging.

#### Lessons for Our Exhibit:

- Incorporate a high level of interactivity and user control over the exhibit components.
- Design modular elements that can be easily modified and reassembled by visitors.
- Provide educational content that supports and enhances the hands-on activities.

## Museum of Science in Boston

### Exhibit: Design Challenges



Figure 3: Program & Collect Design Challenge Exhibit at the Museum of Science in Boston

The Design Challenges exhibit engages visitors in engineering tasks that require problem-solving and critical thinking. Visitors work individually or in groups to design solutions to given challenges. Each challenge presents a specific problem that visitors must solve using provided materials, encouraging innovative thinking and practical application of engineering principles. The exhibit fosters teamwork and communication, as visitors often work together to brainstorm and test solutions. Visitors are encouraged to test their designs, receive feedback, and iterate to improve their solutions, mirroring real-world engineering practices.

#### Lessons for Our Exhibit:

- Integrate problem-solving tasks that challenge visitors to apply their understanding of physics principles.
- Create opportunities for collaborative activities to enhance learning through social interaction.
- Emphasize the iterative design process, encouraging visitors to refine their roller coaster configurations based on trial outcomes.

#### Science Museum of London

##### Exhibit: Launchpad



Figure 4: Launchpad Centre Exhibits at the Science Museum of London

The Launchpad gallery features numerous interactive exhibits that explore fundamental principles of physics, including energy transfer, forces, and motion. It is designed to engage visitors of all ages through hands-on experimentation and observation. Diverse Interactive Stations: Various stations within the gallery focus on different physics concepts, allowing visitors to explore multiple topics in one visit. The exhibits combine visual aids, physical models, and hands-on activities to cater to different learning styles and make abstract concepts tangible. Informational displays and staff educators provide context and explanations, enhancing the educational value of the interactive experiences.

### Lessons for Our Exhibit:

- Design interactive stations within the roller coaster exhibit to cover a range of physics concepts such as potential and kinetic energy, momentum, and friction.
- Use visual aids to complement the hands-on activities, making the concepts more accessible.
- Incorporate educational support materials to guide visitors and enrich their learning experience.

To provide a clear and concise overview of the key strategies and features proposed for the modular roller coaster exhibit, the following table organizes the design principles and lessons learned. Each principle is matched with its source of inspiration and the rationale for its importance in enhancing the educational impact and visitor engagement of the exhibit. This structured approach ensures that best practices from leading science museums are effectively integrated into our exhibit design.

Table 1: Design Principles and Inspirations for the Modular Roller Coaster Exhibit

| Design Principle / Lesson Learned      | Source of Inspiration     | Importance for the Exhibit                                                                                                           |
|----------------------------------------|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Modular roller coaster tracks          | Tinkering Studio          | Encourages experimentation and exploration of engineering principles by allowing visitors to easily assemble and reconfigure tracks. |
| Problem-solving tasks                  | Design Challenges exhibit | Promotes active learning and critical thinking by requiring visitors to design tracks that meet specific criteria.                   |
| Visual aids and informational displays | Launchpad gallery         | Helps visitors understand underlying physics concepts and their application to roller coaster mechanics.                             |
| Iterative design process               | Design Challenges exhibit | Encourages visitors to test, refine, and improve their designs, enhancing their learning experience.                                 |
| Collaboration opportunities            | Design Challenges exhibit | Fosters teamwork and allows visitors to learn from each other's ideas and solutions.                                                 |

By integrating these best practices from leading science museums, the proposed modular roller coaster exhibit aims to provide a rich, engaging, and educational experience for visitors of all ages.

## Literature Review

This literature review encompasses various aspects of interactive exhibit design in science museums, revealing critical insights into methodologies, principles, and best practices that underpin effective educational experiences.

First, hands-on engagement and active learning are essential in informal learning environments. Falk and Dierking's seminal work "The Museum Experience" (2012) highlights these elements' importance. Additionally, research by Griffin and Symington (1997) and Hooper-Greenhill (2000) indicates the

transformative potential of interactive exhibits in fostering visitor curiosity and understanding, underscoring that interactive exhibits serve as powerful tools for facilitating meaningful learning experiences.

Modular exhibit design offers significant benefits such as scalability, adaptability, and cost-effectiveness. Studies by Merriman (2015) and the American Alliance of Museums' publication "Exhibit Makeovers: A Do-It-Yourself Workbook for Small Museums" (2008) provide evidence supporting the incorporation of modular elements into museum exhibit prototypes. These benefits highlight the relevance of a modular approach in exhibit design.

Hands-on learning in physics education effectively enhances student comprehension and retention of scientific concepts. This is demonstrated by studies conducted by Thornton and Sokoloff (1998) and Hake (1998), which support the efficacy of interactive, experiential approaches in education.

Rigorous assessment practices are essential for gauging the effectiveness and usability of exhibit prototypes. Bitgood et al. (1996) and Serrell (1997) outline user testing and evaluation methodologies in museum settings that inform iterative design refinements and improvements. These assessment practices are crucial for developing effective exhibits.

Science museums play a pivotal role in fostering interest and engagement in STEM disciplines among diverse audiences. Publications like "Encouraging Curiosity About Science and Technology Through Everyday Experiences: Interest, Exploration, and Discovery" (National Research Council, 2009) and "Informal Learning and Field Trips" (Falk, 2001) emphasize the integration of STEM education in informal learning environments. Aligning exhibit content with educational standards and curriculum objectives helps museums bridge formal and informal learning contexts.

Digital technologies in museum exhibits have transformative potential in enhancing visitor engagement and learning outcomes. This is highlighted by publications such as "The Museum Experience: Handheld Guides and Other Media" (Anderson, 2008) and "Museums and Digital Culture: New Perspectives and Research" (Marty, 2019). Leveraging digital tools and technologies can create dynamic, personalized, and interactive exhibit experiences catering to diverse visitor preferences and learning styles.

Through an extensive literature review on child safety and accessibility, I identified several crucial factors to consider in the design of the modular roller coaster exhibit. The use of visual cues for puzzle pieces is essential. Research, such as that by Hooper-Greenhill (2000), indicates that clear, color-coded markings significantly aid children in assembling and disassembling components. Incorporating distinct visual cues will enhance the educational value and safety of the exhibit by helping children understand the correct configuration and assembly process, reducing the risk of incorrect assembly that could lead to safety hazards.

Next, the size, density, and weight of the exhibit pieces are critical factors. Studies emphasize that pieces should be large enough to prevent choking hazards, with smooth, rounded edges to avoid injury (Griffin & Symington, 1997). The density and weight should be balanced to ensure that components are light enough for children to handle easily but heavy enough to provide stability and prevent tipping

or accidental displacement during use. This balance is vital to create an exhibit that is both safe and user-friendly for a wide age range.

Additionally, the height of the starting track must be carefully considered. Literature suggests that the track height should be set to a level that is accessible to children while ensuring they can observe the effects of gravity and momentum (Falk & Dierking, 2012). The starting point should be low enough for younger children to reach but high enough to allow for a visible demonstration of potential and kinetic energy. Ensuring the track height is adjustable can cater to different age groups and provide a more inclusive educational experience. By integrating these findings into the exhibit design, I aim to create a safe, accessible, and educational experience for all visitors.

Guiding principles for the exhibit design include the necessity for the installation to be free of electrical outlets, easily repairable, and portable. Ensuring that no outlets are needed not only enhances safety by eliminating electrical hazards but also makes the exhibit more versatile in terms of placement within various venues (Anderson, 2008). Designing the components to be easy to repair means that maintenance can be quickly performed on-site, minimizing downtime and ensuring the exhibit remains functional and safe. Portability is also a key factor, allowing the exhibit to be moved and set up in different locations with ease, broadening its accessibility and reach to various audiences (Bitgood et al., 1996).

Utilizing the insights from these diverse literature sources, the proposed exhibit prototype for the Hands-On Museum is positioned with best practices and innovative methodologies in museum exhibit design. By drawing upon a multidisciplinary knowledge base encompassing interactive design principles, hands-on learning strategies, modular exhibit approaches, user testing methodologies, STEM education integration, and technology trends, the exhibit prototype aims to deliver a transformative and immersive learning experience that inspires curiosity, fosters engagement, and cultivates a lifelong appreciation for science and engineering among museum visitors.

### *1.3 Project Deliverables*

#### Goals and Intended Deliverables

1. The primary goal of the project is to design and develop an engaging interactive exhibit prototype for the Hands-On Museum that effectively educates visitors about physics principles, particularly focusing on potential and kinetic energy, momentum, and friction, while inspiring curiosity in science and engineering. The intended deliverables include:
2. Interactive Exhibit Prototype: The creation of a fully functional prototype of the modular roller coaster exhibit, featuring modular roller coaster tracks, potential and kinetic energy demonstrations, momentum and friction stations, hands-on learning opportunities, and problem-solving challenges.
3. Design Process and Procedure Documentation: A comprehensive documentation outlining the design process and methodology employed throughout the project, including research methodologies, brainstorming sessions, solution evaluation criteria, design calculations, prototyping iterations, and testing methodologies.

4. Alpha and Beta Prototypes: Multiple iterations of alpha prototypes to visualize the design concept and address potential challenges, followed by the development of a refined beta prototype incorporating user feedback and iterative design enhancements.
5. Evaluation and Testing Reports: Reports documenting the results of user testing and evaluation conducted on both alpha and beta prototypes, including observations, feedback analysis, and recommendations for further improvements.
6. Financial, Feasibility, and Relevancy Evaluation: An assessment of the financial feasibility and relevancy of the exhibit prototype, considering factors such as material costs, manufacturing feasibility, maintenance requirements, and alignment with museum's educational goals and visitor demographics.
7. Stakeholder Midway Evaluation: A midway evaluation presentation to museum directors and stakeholders, providing an overview of project progress, prototype demonstrations, and soliciting feedback for refinement.
8. Final Project Evaluation: A final evaluation report assessing the project's success in achieving its goals, including an analysis of the exhibit's educational impact, visitor engagement, and overall effectiveness in meeting project objectives.

The project aims to provide tangible benefits to both the Hands-On Museum and its visitors. For the museum, the exhibit prototype represents a valuable addition to its educational offerings, enhancing its reputation as a premier destination for hands-on learning experiences. The exhibit's interactive and engaging nature is expected to attract a diverse audience, including families, students, and science enthusiasts, thereby increasing museum attendance and revenue streams. Additionally, the exhibit serves as a valuable resource for educators, providing a dynamic platform for teaching STEM concepts in an engaging and memorable way.

#### Quantifiable Measures of Success

1. Visitor Engagement Metrics: Quantitative measures such as visitor traffic, exhibit dwell time, and participation rates in interactive activities will be tracked to assess the level of visitor engagement with the exhibit.
2. Educational Impact Assessment: Surveys and pre/post-visit assessments will be used to measure changes in visitor knowledge, understanding, and interest in physics concepts before and after interacting with the exhibit.
3. Cost and ROI Analysis: A cost-benefit analysis will be conducted to evaluate the return on investment (ROI) of the exhibit prototype, considering factors such as material costs, installation expenses, and projected revenue generation through increased museum attendance.
4. Quality and Usability Evaluation: User feedback and observations collected during testing sessions will be analyzed to identify areas for improvement in exhibit design, functionality, and user experience, with a focus on reducing defects and enhancing overall quality.

## 1.4 Scope

The scope of the project encompasses the design, development, and testing of a modular roller coaster exhibit prototype for the Hands-On Museum. Specifically, the project will focus on the following key areas:

### In Scope

1. **Design and Development:** The project will involve conceptualizing, designing, and prototyping the modular roller coaster exhibit, including the creation of modular roller coaster tracks, potential and kinetic energy demonstrations, momentum and friction stations, hands-on learning opportunities, and problem-solving challenges.
2. **User Testing and Evaluation:** User testing sessions will be conducted to gather feedback and assess the effectiveness of the exhibit prototype in engaging visitors and teaching physics principles. Feedback from visitors will be used to iterate and refine the exhibit design.
3. **Pilot Implementation and Evaluation:** A pilot implementation of the exhibit will be conducted in a real museum setting. This will involve gathering data and conducting research to evaluate the initial success of the exhibit based on two quantitative measures of success: visitor engagement levels and learning outcomes related to physics principles. The findings will inform further design modifications and improvements.
4. **Documentation and Reporting:** Comprehensive documentation will be produced throughout the project, detailing the design process, prototyping iterations, testing methodologies, evaluation results, and recommendations for further improvements. Regular progress reports and presentations will be made to stakeholders to keep them informed of project developments.
5. **Financial and Feasibility Assessment:** An evaluation of the financial feasibility and relevancy of the exhibit prototype will be conducted, considering factors such as material costs, manufacturing feasibility, maintenance requirements, and alignment with museum's educational goals and visitor demographics.

### Out of Scope

1. **Full-Scale Production:** The project will not involve the full-scale production and installation of the exhibit in the museum. While the final exhibit design will be informed by considerations of manufacturability and maintenance, the actual production and installation will be handled by the museum's engineering team after the completion of the project.
2. **Long-Term Maintenance and Operations:** Long-term maintenance and operational considerations beyond the prototyping phase, such as staffing requirements, exhibit upkeep, and visitor management, are outside the scope of this project.
3. **Marketing and Promotion:** While the project aims to create an engaging and educational exhibit, marketing and promotional activities to attract visitors to the museum and the exhibit specifically are not included in the scope. These activities will be the responsibility of the museum's marketing department.
4. **Changes to Museum Infrastructure:** The project will not involve any modifications or changes to the existing infrastructure of the Hands-On Museum. The exhibit prototype will be designed to integrate seamlessly within the museum's existing layout and facilities.

## 1.5 Project Timing

1. Identification of Problem (March): Utilize observational techniques, surveys, and experiments to pinpoint a significant need for a specific type of exhibit within the museum.
2. Development of Design Process and Procedure (April): Establish a structured design process and methodology for the project, incorporating research methodologies, brainstorming sessions, solution evaluation criteria, and design calculations.
3. Research and Information Gathering (early May): Conduct in-depth research to collect relevant information related to exhibit design and construction, including competitor analysis and creating a market value proposition.
4. Brainstorming (mid-May): Facilitate collaborative sessions with the museum team to generate a wide array of potential exhibit solutions, using diverse brainstorming methods such as mind mapping.
5. Alpha Prototyping (May-June): Develop multiple iterations of alpha prototypes to visualize the design concept and address potential challenges, incorporating user feedback and iterative design refinements.
6. Solution Evaluation and Selection (June): Evaluate and select the most feasible exhibit prototype based on criteria such as cost, manufacturability, maintenance, and impact, utilizing decision matrices, user personas, and functional decomposition.
7. Design and Calculations (June): Translate the chosen exhibit prototype into a detailed design, incorporating necessary calculations to ensure feasibility and efficiency.
8. Financial, Feasibility, and Relevancy Evaluation (late June): Assess the financial feasibility and relevancy of the exhibit prototype, considering factors such as material costs, manufacturing feasibility, and alignment with museum's educational goals.
9. Stakeholder Midway Evaluation (late June): Present project progress, prototype demonstrations, and solicit feedback from museum directors and stakeholders for refinement.
10. Beta Prototyping (Early July): Develop a refined beta prototype incorporating feedback from alpha testing, focusing on user experience improvements and design optimizations.
11. Testing and Validation (July): Implement rigorous testing methodologies to validate the functionality and effectiveness of the beta prototype, incorporating user feedback and making necessary adjustments to enhance performance.
12. Final Evaluation (July): Assess the project's success in achieving its goals, including an analysis of the exhibit's educational impact, visitor engagement, and overall effectiveness.

This timeline provides a structured framework for the project, ensuring that key tasks are completed in a logical sequence and allowing for iterative refinement based on user feedback and testing results.

## 2.0 Assumptions and Methodology

In conducting the capstone project for the Hands-On Museum exhibit prototype, several assumptions have been made to guide the design process and ensure the project's feasibility and success. One key assumption is that the target audience, which includes visitors ranging from kindergarten-aged children to adults, has a diverse range of prior knowledge and varying levels of interest in physics principles. This assumption necessitates the design of an exhibit that is not only educational but also highly engaging and accessible to individuals with different learning styles and cognitive abilities. Consequently, the exhibit's interactive elements, such as modular roller coaster tracks and hands-on learning stations, are designed to be intuitive and provide immediate feedback, making complex physics concepts like potential and kinetic energy, momentum, and friction easily understandable and enjoyable for all visitors.

Another critical assumption is the availability and reliability of resources, including financial support from the museum, materials for prototyping, and access to user testing groups. It is assumed that the museum will provide the necessary funding for materials and that there will be enough visitors willing to participate in user testing sessions throughout the project's duration. This assumption is crucial for the iterative design process, as continuous feedback from real users is essential for refining the prototype and ensuring its educational effectiveness. Additionally, the project assumes that the collaborative efforts with the museum staff will be effective, with open communication channels facilitating the exchange of ideas and feedback. This collaborative dynamic is expected to contribute significantly to the project's success by integrating diverse perspectives and expertise into the design and development process.

### Weekly Meetings with the Exhibits Team

#### Regular Updates and Feedback

Weekly meetings with the museum's exhibits team play a crucial role in the development and implementation of the roller coaster exhibit. These sessions are designed to provide updates on the progress of the capstone project, discuss milestones achieved, and address any challenges. Each week, detailed reports are presented to keep all stakeholders informed and aligned with the project's goals. The exhibits team offers valuable feedback based on their expertise, helping to refine the prototype and enhance its educational impact. This iterative process ensures continuous improvement and adherence to the museum's high standards of quality. These meetings also serve as an opportunity for knowledge sharing. The exhibits team shares insights from other projects, providing lessons that can be applied to the roller coaster exhibit. Learning about successful strategies and design principles from other exhibits informs decisions, enhancing the overall visitor experience. Understanding what works well in other contexts aids in incorporating successful strategies and avoid potential pitfalls.

#### Collaborative Problem Solving

The collaborative environment of the weekly meetings fosters collective problem-solving. When challenges arise, the combined expertise of the exhibits teams is leveraged to find effective solutions. This approach accelerates project progress and ensures high-quality outcomes. For instance, if there

are issues with the prototype's structural integrity, the exhibits team can provide proven solutions from their experience with similar problems. Overall, these weekly meetings are vital for building a cohesive and well-integrated exhibit. Regular communication, feedback incorporation, and learning from other projects ensure that the roller coaster exhibit is not only structurally sound and educationally effective but also engaging for visitors. This collaborative approach ensures the final exhibit will be a valuable addition to the Hands-On Museum, offering a memorable learning experience for all visitors.

## 3.0 Design Process

### Initial Prototype Development

The project began with the use of the "Thames and Kosmos Roller Coaster Engineering STEM Experience Kit" to create an initial prototype. This kit was chosen for its ability to provide a basic understanding of the dynamics involved in potential and kinetic energy in roller coasters. The initial phase involved selecting materials from the kit and assembling the plastic roller coaster toy set according to the provided instructions. Various configurations were tested to observe the flexibility and limitations of the track elements and structure. This hands-on approach allowed for a practical evaluation of the kit's potential as an educational tool, highlighting its strengths and weaknesses in real-world applications.

Through the exercise of using the "Thames and Kosmos Roller Coaster Engineering STEM Experience Kit" to create an initial prototype, I gained several key insights that shaped my subsequent design work. I obtained a practical understanding of the dynamics of potential and kinetic energy, which highlighted how these principles could be effectively demonstrated in an educational exhibit. I observed the flexibility and limitations of the kit's materials, informing my decisions on selecting durable, user-friendly materials. Evaluating the kit's educational potential revealed features that enhanced learning and identified areas needing more interactive, problem-solving elements. Additionally, feedback on user experience emphasized the importance of intuitive assembly processes and engaging features. Finally, the exercise underscored the value of an iterative design approach, where prototypes are continuously refined based on user feedback and testing. These insights were instrumental in developing a modular roller coaster exhibit that is both educational and engaging for visitors.

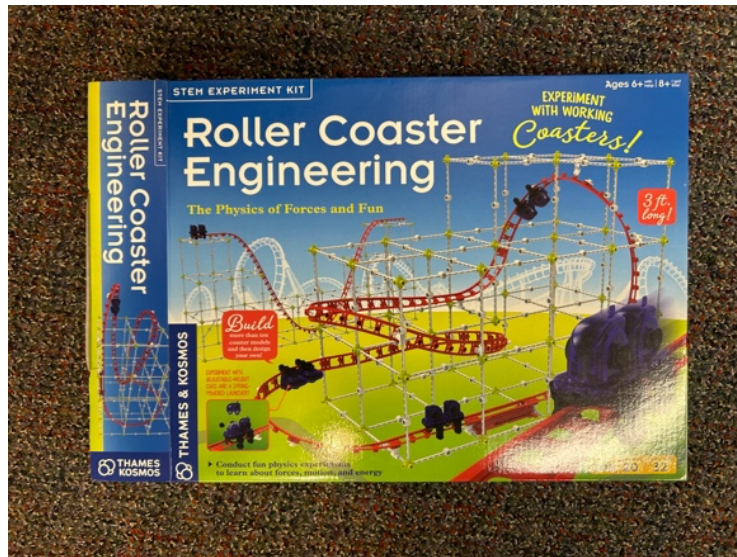


Figure 5: Thames and Kosmos Roller Coaster Engineering STEM Experience Kit

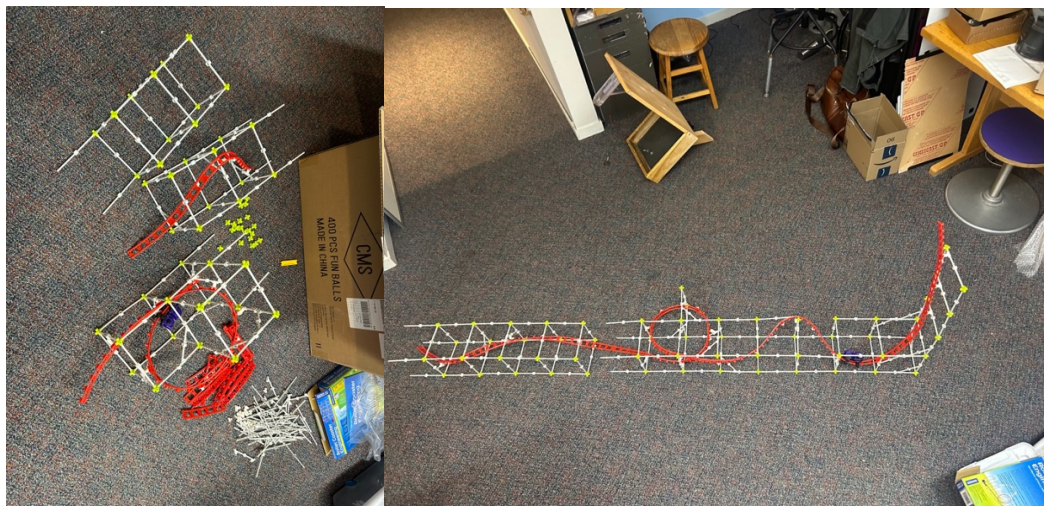


Figure 6 A, B: Thames and Kosmos Roller Coaster Pieces (A) and Construction (B)

### Initial Sketch and Concept Drawing with Sticky Notes

To conceptualize the modular roller coaster design, I began with a hands-on approach using sticky notes. Each sticky note represented a different element of the coaster, such as loops, drops, and turns. This method allowed for a flexible and interactive way to visualize the layout. By moving the sticky notes around on a large surface, we could easily experiment with different configurations and see how various components fit together. This approach highlighted the modular nature of the design, enabling the exploration multiple design possibilities and quickly iterate on ideas. The sticky notes facilitated collaborative brainstorming sessions, where museum exhibits staff members could contribute and adjust elements in real-time, fostering a dynamic and creative design process.

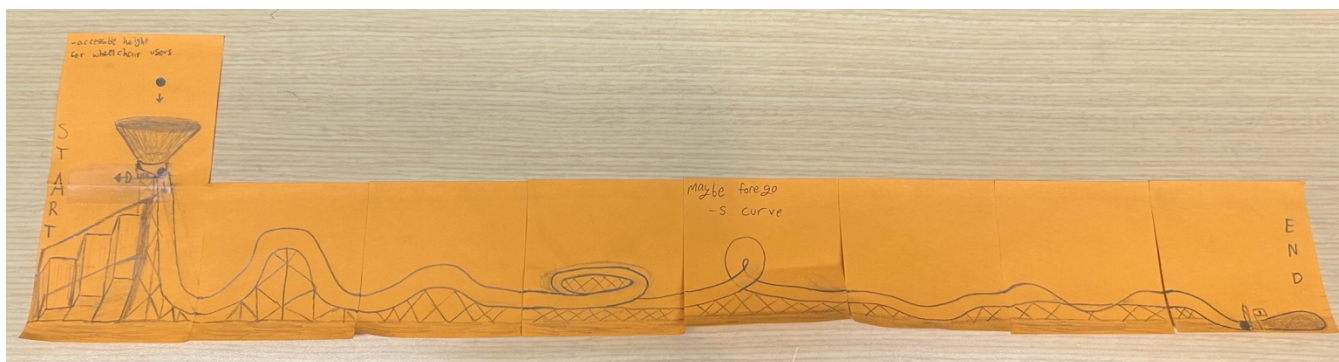


Figure 7: Sticky Note Concept Sketch

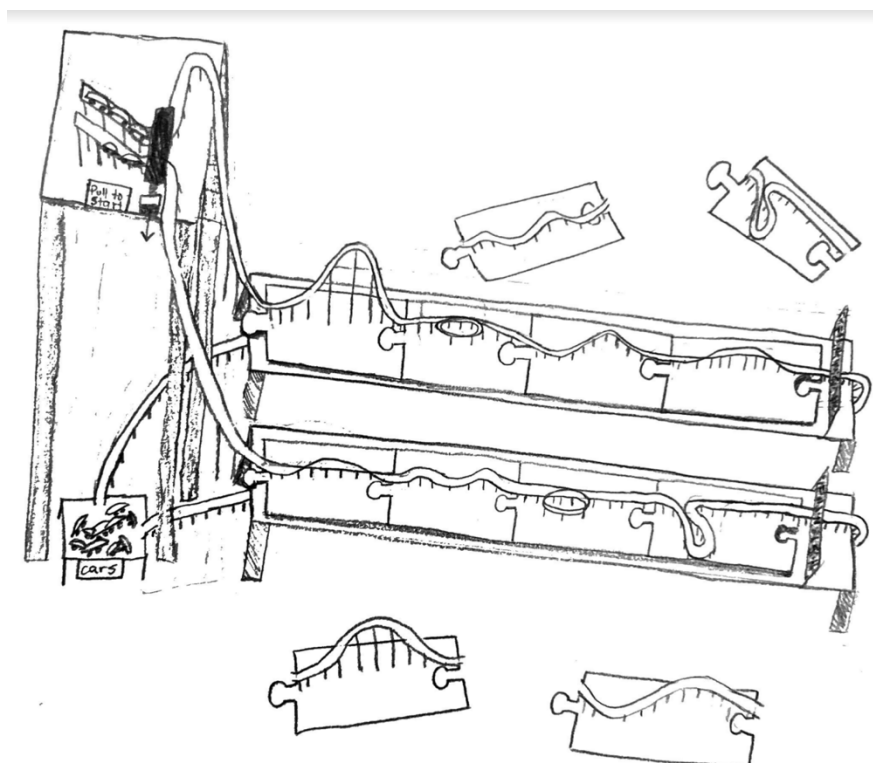


Figure 8: Modular Racing Coaster Exhibit Concept Design

### Collaborative Conceptualization and Design Prioritization

The initial sketch and concept drawing using sticky notes provided a rich collaborative and creative process that led to significant learnings and priority setting for the modular roller coaster exhibit. Engaging in this hands-on approach allowed me to visualize the layout dynamically and experiment with different configurations in a flexible manner. The sticky notes represented various elements of the coaster, such as loops, drops, and turns, and could be easily repositioned to explore multiple design possibilities.

Through this collaborative process, several key insights and design priorities emerged:

- **Modular Flexibility:** The ability to move sticky notes around highlighted the importance of modularity in the exhibit design. It became clear that the modular elements should be easily adjustable to allow for a variety of track configurations, fostering creativity, experimentation, and iterative learning among visitors.
- **Interactive Engagement:** The tactile nature of using sticky notes emphasized the value of hands-on interaction in the final exhibit. Elements such as adjustable tracks and interactive stations became top priorities to ensure that visitors could actively engage with the exhibit, rather than passively observe.
- **Collaboration and Inclusion:** The collaborative brainstorming sessions with museum exhibit staff members underscored the importance of creating an exhibit that encourages teamwork and idea-sharing among visitors. Features that promote collaboration, such as team-based problem-solving tasks and shared assembly challenges, were prioritized to replicate the dynamic and inclusive creative process experienced during the design phase.
- **Educational Focus:** Discussions and real-time adjustments during the sketching sessions clarified which physics principles should be prominently featured. Elements that clearly demonstrated potential and kinetic energy, momentum, and friction were prioritized to enhance the educational value of the exhibit.
- **Practical Considerations:** The interactive sketching process helped identify practical considerations, such as spatial constraints and visitor flow. Prioritizing elements that fit well within the exhibit space and facilitated smooth visitor movement became essential to ensure a seamless and enjoyable experience.
- **Iterative Design:** The ease of modifying the sticky note layout underscored the necessity of an iterative design process in the exhibit's development. Continuous testing and refining of exhibit elements based on user feedback were incorporated as crucial steps to ensure the final design met educational and engagement goals.

Overall, the collaborative and interactive nature of using sticky notes for initial sketches provided valuable insights into designing an engaging, educational, and highly modular roller coaster exhibit. Prioritizing modular flexibility, interactive engagement, collaboration, educational focus, practical layout, and iterative design processes were key outcomes from this creative exercise.

### **Building a Prototype with a Wooden Base**

After evaluating the initial prototype constructed from a plastic roller coaster toy kit, I realized that the toy lacked the necessary structural integrity and design flexibility for my exhibit. The plastic components were too flimsy, limiting the types of elements I could create and compromising the stability required for an interactive exhibit. To address these issues, I decided to build a more robust prototype using a wooden base. Utilizing the resources available at the Hands-On Museum wood shop, I constructed a sturdy wooden frame that provided a solid foundation for the track. This wooden base significantly enhanced the structural stability, allowing me to securely integrate the plastic track segments from the initial toy kit. By combining the durability of the wooden frame with the versatility of the plastic tracks, I was able to create a prototype that supported various educational and interactive elements, ensuring a more stable and engaging experience for visitors.

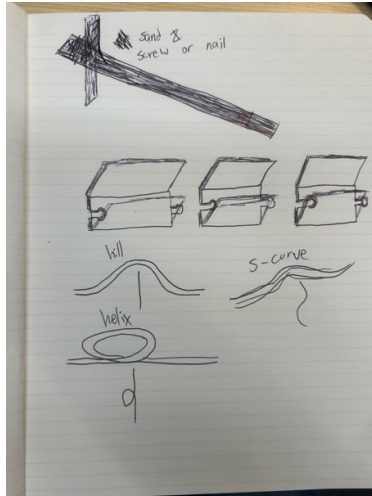


Figure 9: Wooden Prototype Concept Sketch



Figure 10: Construction of Wooden Puzzle Pieces as Track Element Bases

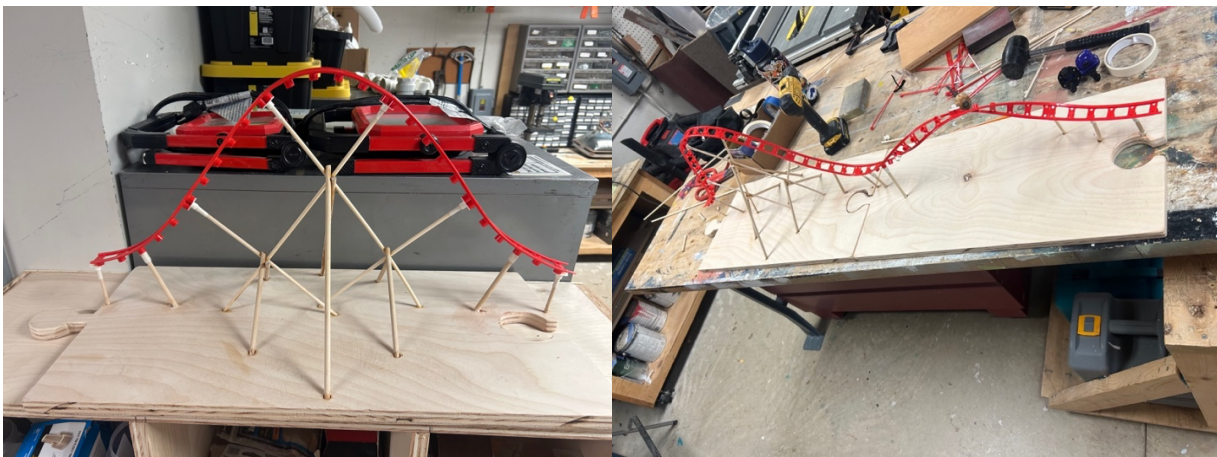


Figure 11: Assembly of Wooden Modular Coaster Prototype

### Evaluation and Iteration

Following the initial prototype's assembly and testing, a comprehensive assessment was conducted. It was found that the initial prototype was too structurally flexible and creatively restrictive, limiting the

types of elements that could be created. This prior realization led to a critical decision to enhance the structural integrity of the prototype. The wooden frame would provide the necessary stability. Utilizing the resources available at the Hands-On Museum wood shop, a sturdy wooden frame was constructed. This involved precise cutting, assembly, and securing of wooden puzzle-piece shaped pieces to form a stable base for the track. The plastic track segments from the initial toy set were then integrated into the wooden frame, allowing for more stable and diverse configurations. This iterative approach ensured that the prototype could support various educational and interactive elements effectively.



Figure 12: Prototype Created for User Testing

### Measurement System

Key performance metrics were established to guide the development and evaluation of the prototype. These metrics included structural stability, ease of assembly, educational effectiveness, and user engagement. Observational data were collected by museum exhibits staff members while I assembled the prototype. Physical load testing was done to evaluate the structural integrity and flexibility of different configurations. Staff members attempted to reconfigure the track pieces and apply pressure to the prototype in various places to validate durability. Feedback was collected from museum visitors on the ease of use and educational value of the prototype. This measurement approach ensured that the prototype met the educational and engagement goals of the project.

**Virtual Design with NoLimits 2** Following the initial sketching phase, I utilized NoLimits 2, a professional roller coaster simulation software, to create a virtual model of the exhibit. This program allowed me to design the coaster with precision, incorporating detailed elements such as track curvature, elevation changes, and speed variations. NoLimits 2 provided a comprehensive platform to test the feasibility of our design concepts, enabling us to simulate the physics of potential and kinetic energy, momentum, and friction in a virtual environment. The software's advanced features allowed for realistic modeling and visualization, giving valuable insights into the ride dynamics and structural integrity. By using NoLimits 2, I could refine our design, optimize the track layout, and ensure that the modular components would provide a safe and engaging experience for museum visitors.

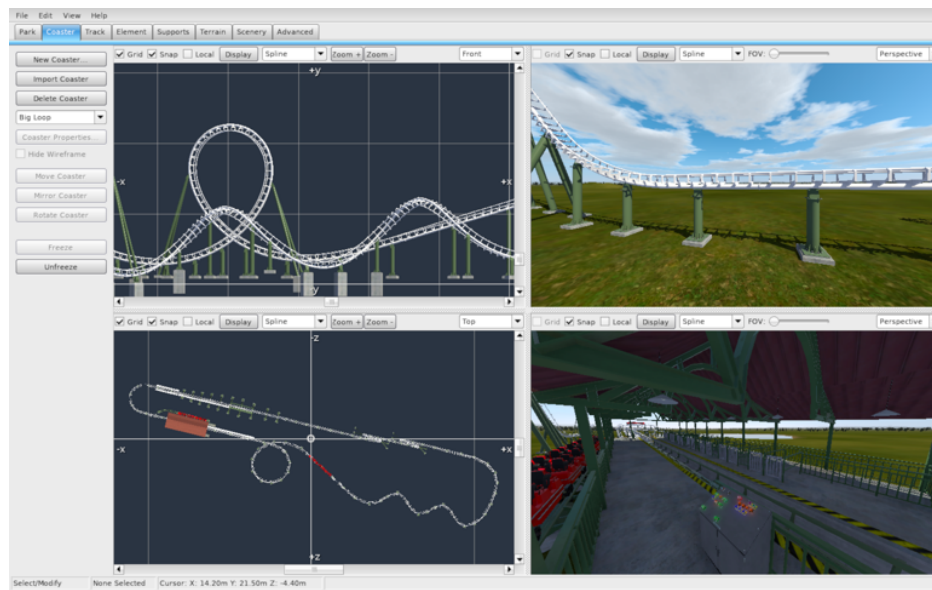


Figure 13: NoLimits 2 Coaster Editing Interface

### Elements Explored in NoLimits 2:

**Track Layout:** NoLimits 2 allows for precise design of the modular track pieces, including curves, slopes, and straight sections. This virtual environment lets me experiment with different configurations to create an engaging and educational experience. Factors like length, complexity, and potential for showcasing physics principles will guide my layout choices.

**Elevation Changes:** The software allows for simulating different heights and drops within the modular system. This lets me test the impact of potential and kinetic energy transfer as the coaster car travels throughout the track. I can explore features like a significant starting hill to provide initial momentum, followed by dips and climbs to demonstrate energy conversion.

**Speed Variations:** By simulating the physics within NoLimits 2, I can analyze the impact of different track configurations on the coaster's speed. This helps ensure a safe and thrilling ride experience for museum visitors. I can test elements like loops, corkscrews, and controlled descents to create a dynamic and exciting ride while maintaining safety limitations.

**Learning from the Virtual Design Process:**

**Feasibility Testing:** NoLimits 2 allows me to identify any physical limitations or unrealistic elements in the initial design concepts. By testing the virtual model, I can refine the design to ensure the modular components function smoothly and safely in a real-world setting.

**Physics Visualization:** The virtual environment allows for a clear visualization of how the coaster interacts with physics principles. This helps me identify opportunities to showcase concepts like momentum transfer, energy conversion, and friction within the exhibit. I can use this information to design informative displays or challenges that highlight these physics principles for museum visitors.

By utilizing NoLimits 2 for virtual design, I gain valuable insights that will guide the final design choices. The ability to test feasibility, visualize physics principles, and ensure a safe and engaging experience in a virtual environment contributes to a more successful and educational exhibit for the museum.



Figure 14: NoLimits 2 Modular Coaster Prototype Ride View

### Testing Methods

To validate the design and functionality of the prototype, a combination of virtual and physical testing methods was employed. The NoLimits roller coaster simulation software was used to model various track designs and configurations virtually alongside physical prototyping. This software allowed for detailed analysis of potential and kinetic energy, momentum, and friction in different track designs. Multiple simulations were run to identify optimal track designs and configurations that would be

feasible and effective for the physical prototype. For physical testing, both the initial plastic roller coaster toy set and the enhanced wooden frame with integrated plastic tracks were tested in the Hands-On Museum. Various configurations were assembled, and their performance was evaluated based on structural stability, educational value, and user interaction.

## Data Collection and Validation Methodology

### Sampling Strategy

The prototype was tested by museum visitors for operational feedback and engagement observation. A random sampling strategy was implemented to gather comprehensive feedback from museum visitors. Visitors of various ages, including children, teenagers, and adults, were invited to interact with the prototypes during designated testing sessions. This approach ensured that feedback was collected from a wide demographic range. Structured observation and feedback discussions were used to collect data from participants, focusing on ease of use, engagement level, and understanding of the physics principles demonstrated by the exhibit. This strategy ensured that the prototype was evaluated from multiple perspectives, providing valuable insights for further refinement.

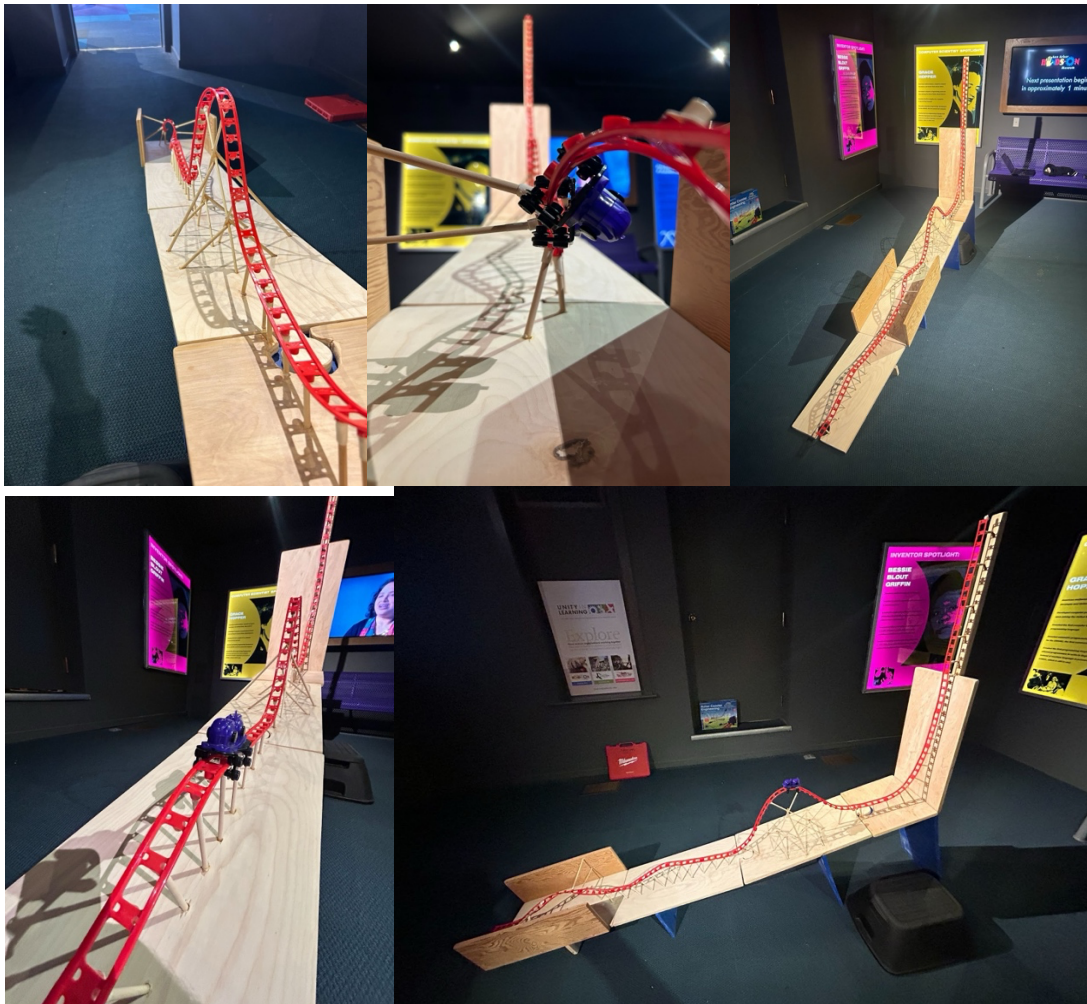


Figure 15: Modular Roller Coaster Physical Prototype for User Testing

### **Prototype Setup and Data Collection**

The exhibit prototype was set up in the Hands-On Museum Theater of Design, ensuring a controlled environment for consistent data collection. The wooden frame and plastic track segments, along with additional educational displays and interactive elements, were arranged for visitors to interact with. Observations were made on how visitors engaged with the prototypes, including time spent on each configuration and the number of successful roller coaster runs. Participants were asked to respond to questions rating their experience, understanding of the physics principles, and any suggestions for improvement. Additionally, interactions were observed with the help of education staff member at the museum to capture detailed behavior and engagement. This thorough approach to data collection ensured that all aspects of the prototype's performance were evaluated comprehensively.

### **Validation**

The most important principles guiding the testing and design process are education, engagement, and safety. Ensuring that the exhibit provided an educational experience was paramount; the virtual simulations and physical prototypes were designed to clearly demonstrate fundamental principles of physics such as energy transformation and motion. Engagement was equally critical, as the goal was to create an interactive and captivating experience for children. The modular nature of the design, combined with the ability to rearrange and experiment with different track configurations, was crucial in maintaining high levels of user engagement. Safety was a foundational principle throughout all testing phases, ensuring that all components were sturdy, free of sharp edges, and appropriately scaled for children to handle. By adhering to these principles, the final design aimed to deliver a comprehensive, interactive, and safe educational exhibit. Validation through rigorous user testing affirms the successful integration of these guiding principles into the exhibit design. Children participating in the user testing sessions demonstrated significant educational gains, showing improved understanding of physics concepts through hands-on interaction with the exhibit's modular components. Their engagement was moderately high, with enthusiastic exploration and experimentation observed across different age groups. Observations and feedback from educators and parents highlight the exhibit's effectiveness in fostering curiosity and sustained interest in STEM subjects. Safety protocols were evaluated during testing, ensuring that all aspects of the exhibit met stringent safety standards without compromising the interactive and educational experience. Overall, user testing provides empirical validation that the exhibit design effectively balanced education, engagement, and safety, meeting the intended goals of enriching learning experiences in a safe and enjoyable manner.

# 4.0 Prototype Performance and User Feedback

## Educational Impact

One of the key metrics was the educational impact of the exhibit. Visitor feedback and observational data indicated a high level of engagement and understanding of the principles of potential and kinetic energy, momentum, and friction.

## User Engagement

User engagement was measured through the duration of interaction and the number of different configurations attempted by visitors. Table 2 illustrates the average time visitors spent interacting with the exhibit and the average number of track configurations they tried.

Table 2: Engagement Metrics During User Testing

| Metric                                     | Description                                           | Measurement Method                                              | Improvement Potential                                                                |
|--------------------------------------------|-------------------------------------------------------|-----------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Task Completion Rate                       | 70% of users who successfully build a working coaster | Recorded data during testing                                    | Increase with improved track design, instructions                                    |
| Time to Complete Task                      | Average 3 min 18 seconds                              | Tracked time for each user, calculated average                  | Decrease with improved usability and clarity                                         |
| User Satisfaction Score                    | 4.5/5                                                 | Administer brief verbal survey after testing, calculate average | Increase with more engaging and educational experience, less bias in survey delivery |
| Number of Unique Track Configurations Used | 6                                                     | Observed and recorded during testing                            | Increase by creating additional coaster elements                                     |

The data indicates that visitors spent an average of 3 minutes and 18 seconds interacting with the exhibit and attempted six different configurations. This high level of engagement is indicative of the exhibit’s success in capturing and maintaining visitor interest.

## Problems Encountered and Solutions

Throughout the project, several problems were encountered and addressed. The initial plastic prototype's lack of stability was a major issue. This was resolved by constructing a more durable wooden frame, as previously discussed. Another challenge was ensuring the track pieces were interchangeable and easy to assemble. Iterative testing and feedback led to refinements in the track design, making it more user-friendly and intuitive.

Additionally, sourcing materials and coordinating with local manufacturers for injection molding poses logistical challenges. These were mitigated by establishing clear communication channels and timelines with the partners, ensuring that production stayed on schedule.

## **Significance and Implications**

The findings from this project have significant implications for the Hands-On Museum and similar educational institutions. The success of the roller coaster exhibit in engaging and educating visitors suggests that interactive, hands-on exhibits are effective tools for teaching complex scientific principles. The iterative design process and collaboration with local partners also provide a model for future exhibit development projects, emphasizing the importance of user feedback and structural integrity.

The exhibit's high engagement and educational value indicate its potential to attract and retain museum visitors, contributing to increased attendance and visitor satisfaction. Furthermore, the project showcases the feasibility of using local resources and partnerships to enhance educational offerings, fostering community involvement and support.

## **Exhibit Proposal for Grant Funding**

### **Exhibit Objectives**

- **Project Goal:** Design and develop an engaging, interactive exhibit prototype for the Hands-On Museum that educates visitors about fundamental physics principles, particularly focusing on energy, momentum, and inertia.
- **Problem Addressed:** The need for a new, interactive exhibit that simplifies complex physics mechanics concepts for learners of all ages.
- **Interactive Exhibit:** A functional exhibit featuring modular roller coaster tracks, energy, momentum, and inertia information display, hands-on learning, problem-solving, and competitive challenges.

### **Approach**

- Modular exhibit design made for easy assembly and reconfiguration of components.
- Problem-solving tasks to challenge visitors to apply their understanding of physics principles.
- Educational content to support and enhance the hands-on activities and learning experience.
- Multiple prototype iterations with user testing and feedback for continuous improvement.
- Iterative design process with rigorous user testing incorporated and multiple prototypes.

### **Expected Benefits**

- **Enhanced Learning:** Improved visitor understanding of physics mechanics.
- **Increased Engagement:** Interactive nature of the exhibit promotes engagement.
- **Educational Support:** Physical resource for educators to teach STEM concepts.
- **Visitor Attraction:** Potential to attract new visitors and boost museum revenue.

### **If a grant is awarded:**

- **Exhibit Development:** Grant funds will be used for lab and material to manufacture the modular roller coaster tracks, energy, momentum, and inertia display information. The functional exhibit will then be constructed based on the user-tested design.

- Educational Content Creation: Clear and engaging signage explaining key physics concepts (potential & kinetic energy, momentum, inertia) will be developed. Interactive elements will be designed to encourage problem-solving and experimentation throughout the exhibit.
- Museum Integration: Collaboration with the Hands-On Museum team will ensure smooth visitor flow and successful addition of the exhibit into the space. Training materials will be developed for museum staff on exhibit operation and visitor interaction.

| The performance expectation above was developed using the following elements from the NRC document <i>A Framework for K-12 Science Education</i> :                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Science and Engineering Practices</b><br><b>Analyzing and Interpreting Data</b><br>Analyzing data in 6–8 builds on K–5 and progresses to extending quantitative analysis to investigations, distinguishing between correlation and causation, and basic statistical techniques of data and error analysis. <ul style="list-style-type: none"> <li>• Construct and interpret graphical displays of data to identify linear and nonlinear relationships.</li> </ul> | <b>Disciplinary Core Ideas</b><br><b>PS3.A: Definitions of Energy</b> <ul style="list-style-type: none"> <li>• Motion energy is properly called kinetic energy; it is proportional to the mass of the moving object and grows with the square of its speed.</li> </ul> | <b>Crosscutting Concepts</b><br><b>Scale, Proportion, and Quantity</b> <ul style="list-style-type: none"> <li>• Proportional relationships (e.g. speed as the ratio of distance traveled to time taken) among different types of quantities provide information about the magnitude of properties and processes.</li> </ul> |

Figure 16: NRC K-12 Science Education Motion and Energy Framework

## 5.0 Recommendations

### Transitioning from Final Prototype to Actual Exhibit

#### Partnership and Collaboration

As the project transitions from the final prototype phase to the creation of the actual exhibit, collaboration with local companies and stakeholders becomes essential. The Hands-On Museum will leverage partnerships with local manufacturers and businesses to ensure high-quality production and implementation of the exhibit. Key local companies specializing in injection molding and signage production will be engaged to bring the prototype to life. This collaborative effort will not only support local businesses but also ensure that the exhibit meets professional standards of durability and aesthetic appeal.

#### Injection Molding of Parts

The first step in this transition involves working with local companies that specialize in injection molding to produce the various components of the roller coaster track. Injection molding is chosen for its ability to create precise, durable, and consistent parts, which are crucial for a hands-on exhibit that will be used by numerous visitors. The design specifications developed during the prototyping phase will be shared with the manufacturing partners. These partners will use these designs to create molds and produce the final parts. Regular consultations and quality checks will be conducted to ensure that the produced components match the intended design and functionality.

#### Creation of Official Signage

Simultaneously, the development of official signage will be undertaken to enhance the educational and visual appeal of the exhibit. Local companies specializing in educational and interactive exhibit signage will be engaged to design and produce these elements. The signage will include detailed explanations

of the physics principles demonstrated by the exhibit, such as potential and kinetic energy, momentum, and friction. It will also provide instructions and challenges to encourage visitors to engage actively with the exhibit. Design considerations will ensure that the signage and written explanations of mechanics principles are accessible and engaging to a diverse audience, including young children and adults.

### **Installation and Testing**

Once the parts and signage are produced, the installation phase will commence in the fall, and the museum's engineering team will assemble the final exhibit. This phase will involve rigorous testing to ensure that all components function as intended and that the exhibit provides a safe and engaging experience for visitors. Adjustments and refinements will be made based on testing outcomes to address any issues that arise during assembly.

### **Educational Impact and Community Engagement**

The successful transition from prototype to exhibit will significantly enhance the museum's offerings, providing a robust educational tool that fosters curiosity and learning in physics and engineering. Additionally, this project will highlight the value of community engagement and local partnerships in achieving educational and cultural goals. The exhibit is expected to attract increased attendance and foster a deeper appreciation for science and engineering among visitors. By partnering with local companies for injection molding and signage production, and by conducting thorough testing and refinement, the project aims to deliver a high-quality, engaging, and educational exhibit. This collaborative approach ensures that the exhibit will be a valuable and lasting addition to the Hands-On Museum, benefitting the community for years to come.

## **6.0 Impact/Financial Benefits**

### **Impact of Adopting Recommendations**

Adopting the recommendations for the modular roller coaster exhibit is expected to have a significant positive impact on both educational engagement and the financial viability of the Hands-On Museum. The proposed exhibit, inspired by successful models from other science museums, integrates interactive and collaborative elements while setting up the museum to succeed financially.

### **Quantitative Financial Benefits**

From a financial perspective, the project is projected to yield substantial benefits. The initial cost to manufacture the modular roller coaster exhibits, including printing plastic parts, creating signage, and staffing, is approximately \$15,000 per exhibit or \$6,000 without staffing. By producing multiple copies of the exhibit and selling them to other museums at a price of \$20,000 each, the Hands-On Museum stands to generate significant revenue. For each exhibit sold, the museum would achieve a gross profit of \$14,000. Assuming the museum can sell at least five exhibits annually, this would result in a total annual revenue of \$100,000 and a gross profit of \$70,000. This revenue can be reinvested into the museum to fund new exhibits, enhance existing ones, or support other educational initiatives.

## **Ethical Context and Unintended Consequences**

Ethically, the project aligns with the museum's mission to provide high-quality educational experiences. However, it is essential to consider potential unintended consequences. For example, the focus on profitability should not overshadow the primary educational goals. There is a risk that in prioritizing financial gains, the museum might compromise on the quality or educational integrity of the exhibits. Additionally, while the exhibit is designed to be engaging and informative, it must also be accessible to a diverse audience. Care should be taken to ensure that the exhibits do not inadvertently exclude any groups due to complex language, high cost, or physical accessibility barriers.

## **Estimation Method and Scenarios of Cost Savings**

The estimation of financial benefits is based on a straightforward cost-revenue analysis. The manufacturing cost of \$6,000 per exhibit includes material costs and overhead. The selling price of \$20,000 is set competitively to ensure market acceptance while providing a sufficient profit margin. Different scenarios can be considered to understand the range of potential financial outcomes. For instance, if the museum manages to optimize the production process and reduce manufacturing costs by 10%, the profit margin per exhibit would increase, leading to higher overall profitability. Conversely, if market conditions require reducing the selling price to \$18,000 to remain competitive, the profit margin would decrease, but the volume of sales could potentially increase, compensating for the lower unit profit.

By presenting these quantitative financial, the project has the potential to enhance the museum's financial sustainability while advancing its educational mission. The careful balance between financial viability and educational integrity ensures that the project not only supports the museum's growth but also its commitment to providing enriching and accessible learning experiences for all visitors.

## **7.0 Summary/Conclusions**

The goal of the modular rollercoaster exhibit at the Hands-on Museum was to create an engaging and educational experience for visitors to explore physics through building and experimenting. Initial development focused on offering pre-built tracks with varying complexity. User testing revealed high engagement with rearranging track pieces. The exhibit prioritizes user experimentation with configurations to create functional coasters. Additional testing identified key challenges: unintuitive track connections, track instability, and difficulty aligning the train. User preferences, such as interest in multi-car trains and variable starting points, suggested a desire for more complexity and challenge.

Analysis of the user testing data led to several recommendations. Final track design requires future work, focusing on intuitive connection pieces, stability through stoppers and high-friction surfaces, and potential visual guides for train alignment. Aesthetics can be enhanced by creating a visually unified track and base and exploring colored track sets for building dual racing coasters. Clear and engaging instructions with visuals are important to guide users towards rearranging the track and understanding physics concepts. Introducing features like attachable cars, variable starting points considering age

appropriateness, and tiered platforms can further enhance the experience. Engaging activities and challenges that encourage exploration and competition with different track configurations will make the exhibit even more appealing.

These improvements aim to increase user engagement by encouraging them to build and experiment. A more user-friendly track design and clear instructions will enhance the overall experience. Ultimately, the exhibit aims to promote learning through exploration and experimentation with physics concepts. The findings can be applied to other modular construction toys or exhibits where user assembly plays a central role. The importance of clear instructions, intuitive design, and engaging activities is likely to be valuable across similar contexts. This development process highlights the effectiveness of iterative design and user testing in creating engaging and educational experiences.

In conclusion, the development process for the modular rollercoaster exhibit at the Hands-On Museum highlights the power of user-centered design. By prioritizing user feedback throughout the design stages, I prototyped the collection of pre-built tracks into a dynamic tool for hands-on exploration of physics principles. The modular track components, clear instructions, and engaging activities will not only provide an engaging user experience but also create an effective learning environment. This user-centered approach, emphasizing clear communication, intuitive design, and engaging activities, can be a valuable roadmap for developing other interactive exhibits and educational tools across various contexts. Furthermore, the modular and easily adaptable nature of the exhibit makes it a beneficial business venture, attracting visitors to the museum and offering opportunities for revenue generation through sales of additional copies.

The AAHOM's modular rollercoaster exhibit will hold the potential to spark curiosity, ignite a passion for STEM education, and leave a lasting impression on visitors of all ages.

## 8.0 Appendices

### Appendix A: User Testing Results

#### Participants:

- 15 Camp Kids (Age range likely 5-10)
- 12 Kids with 8 Parents/Grandparents (Age range likely 3-12 for kids)

#### Observations:

- **Initial Behavior:** All participants, regardless of age group, initially attempted to send the car down the pre-built track without any adjustments or reading instructions. This suggests a strong focus on immediate play and a potential lack of understanding of the core concept (building a functional coaster). Competition should help encourage coaster reconfiguration.
- **Learning Through Play:** Some children were able to learn independently that the large hill needed to be placed at the beginning for sufficient momentum. This indicates the potential for

implicit learning through experimentation. Users would push the train when it didn't complete the track.

- **Track Usability Issues:**
  - **Connection Pieces:** Participants found the track connection pieces to be unintuitive and difficult to use. This is a roadblock to rearranging the track and hindering core gameplay.
  - **Stability:** The lack of stoppers on the bottom and sides of the track pieces and low-friction surfaces caused the pieces to slide, further frustrating users.
  - **Train Alignment:** Participants found it challenging to line up the train correctly onto the track.
- **User Preferences:**
  - **Mid-Track Starting Point:** Some users expressed a desire to start the car in the middle of the track, suggesting an interest in defying expectations or creating more complex courses (maybe older children).
  - **Multi-Car Trains:** Participants showed interest in adding more cars to the train, indicating a desire for additional variables and challenges.
  - **Competition:** Introducing competition could encourage more rearrangement and overall engagement. This aligns with the "The Curious Tinkerer" and "Junior Engineer" personas.

#### Recommendations:

- **Clear Instructions:** Provide clear and engaging instructions, with visuals, to guide users towards rearranging the track and understanding the physics concepts.
- **Improved Track Design:**
  - Redesign connection pieces to be more intuitive and user-friendly.
  - Implement stoppers on the bottom and sides of track pieces and high-friction surfaces to prevent sliding.
  - Consider adding visual guides to aid in train alignment.
  - Add a start mechanism to release trains at the same time for racing.
  - Offer magnetically attachable cars for building longer trains.
  - Consider incorporating elements that allow users to start the car from different points on the track.
- **Engaging Activities:** Introduce activities or challenges that encourage competition and exploration of track configurations. This could cater to users who enjoy problem-solving and competition.

#### Aesthetics:

- **Track Cohesion:** Design the track pieces and base to visually appear as one continuous unit. This can enhance the overall aesthetic and create a seamless playing experience. For racing, each set of pieces should be a different color. 1 color per set.

### Favorite Track Pieces:

- **Big Thrill, Big Attention:** The big hill and corkscrew received more user attention than the small hill. This suggests a preference for more dramatic and exciting track elements.

### Scale and Accessibility:

- **Reaching the Top:** Address the issue of some children (younger) being unable to reach the top of the track, even with a step stool. Consider lowering the overall height or implementing a way for the train to be dropped at any height.
- **Intimidating Pieces:** Some larger pieces might be intimidating for very young children to pick up and move. Explore ways to decrease the size of these pieces slightly while maintaining functionality, or consider offering alternative, smaller versions for younger age groups. Maybe have pieces slide across the floor.

## Appendix B: Modular Roller Coaster Exhibit User Personas

### The Playful Explorer - Preschooler

- **Goal:** Explore the visually appealing pieces and make the train move down the track.
- **Motivation:** Sensory play and immediate cause-and-effect.
- **Skills:** Basic manipulation and hand-eye coordination potentially with some help.
- **Pain Points:** Difficulty aligning the train and track pieces, frustration with the train not going all the way on the first attempt.
- **Engagement:** Enjoys seeing the train go down a pre-built track, might need help aligning pieces.

### The Instant Gratification Kid - Lower Elementary Schooler

- **Goal:** Play with the pre-built track and launch the train immediately.
- **Motivation:** Immediate fun and action.
- **Skills:** Basic understanding of cause-and-effect.
- **Pain Points:** Frustrated by initial failures to get the train onto the track, unintuitive track connections, distracted by other exhibits
- **Engagement:** Wants to start playing right away, might not care about rearranging the track, gets impatient with complex setups, tries it once and leaves.

### The Junior Engineer - Upper Elementary Schooler

- **Goal:** Rearrange the track to create a challenging and exciting course for the car.
- **Motivation:** Creativity and problem-solving.
- **Skills:** Basic understanding of cause-and-effect enjoys following instructions.
- **Pain Points:** Frustrated by initial failures due to physics principles, unwilling to share
- **Engagement:** Enjoys the challenge of rearranging the track based on instructions or pictures, observes how the car reacts to different configurations, takes pride in creating a successful course.

### **The Curious Tinkerer - Middle/High Schooler**

- **Goal:** Experiment with different track configurations to see what works best.
- **Motivation:** Immediate fun, challenge and exploration of cause-and-effect.
- **Skills:** Enjoys experimentation, grasps basic physics concepts.
- **Pain Points:** Limited by the pre-built track pieces, might lose interest without competition.
- **Engagement:** Tries various track arrangements, tests different starting points, enjoys figuring out how to make the train complete the course.

### **The Supportive Coach - Parent/Educator**

- **Goal:** Encourage children to explore the exhibit and experiment with the track.
- **Motivation:** Immediate fun, fostering curiosity and problem-solving skills.
- **Skills:** Clear communication, ability to guide exploration.
- **Pain Points:** Lack of clear instructions on which configurations will get the train to complete the course, not focused enough to rearrange pieces multiple times.
- **Engagement:** Helps rearrange the track pieces, explains basic physics concepts in a fun way, enjoys seeing children learn through play.

### **The Passive Observer - Grandparent**

- **Goal:** Enjoy watching their grandchild play.
- **Motivation:** Spending time with family and observing their grandchild's learning.
- **Skills:** Observation and social interaction.
- **Pain Points:** More interested in grandchildren than the exhibit itself.
- **Engagement:** Watches their grandchild play, may occasionally help with larger track pieces, prefers simpler track layouts.

User testing completed: July 2nd, 2024

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